Universität Konstanz

Introduction to Natural Language Processing (NLP) and Large Language Models (LLMs)

Jun.-Prof. Dr. Miriam Schirmer

Konstanz, 22.10.2026

Outline for Today

1. Topic Intro
2. Admin
3. Questions



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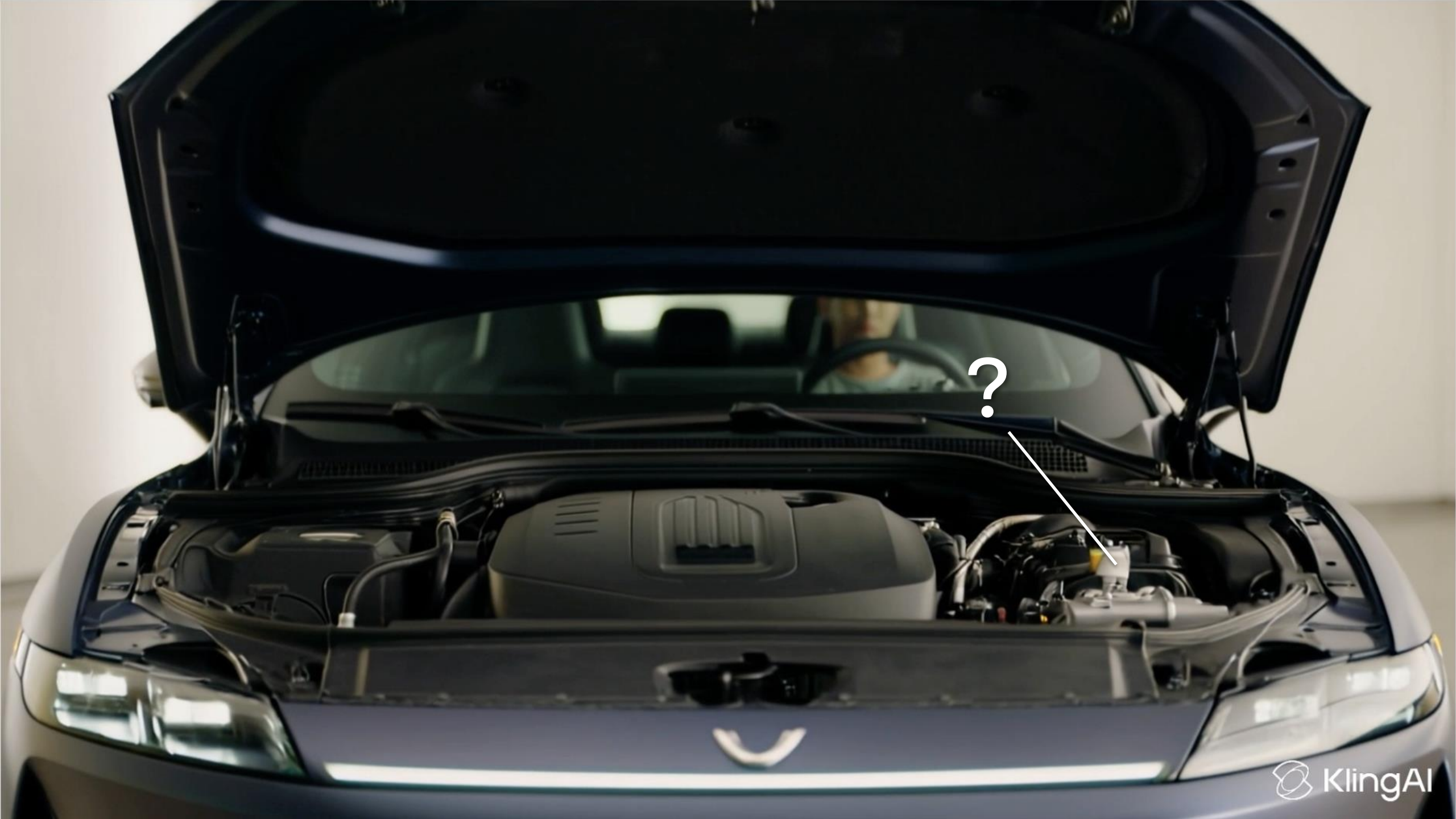


What comes to mind when you hear "language model"?

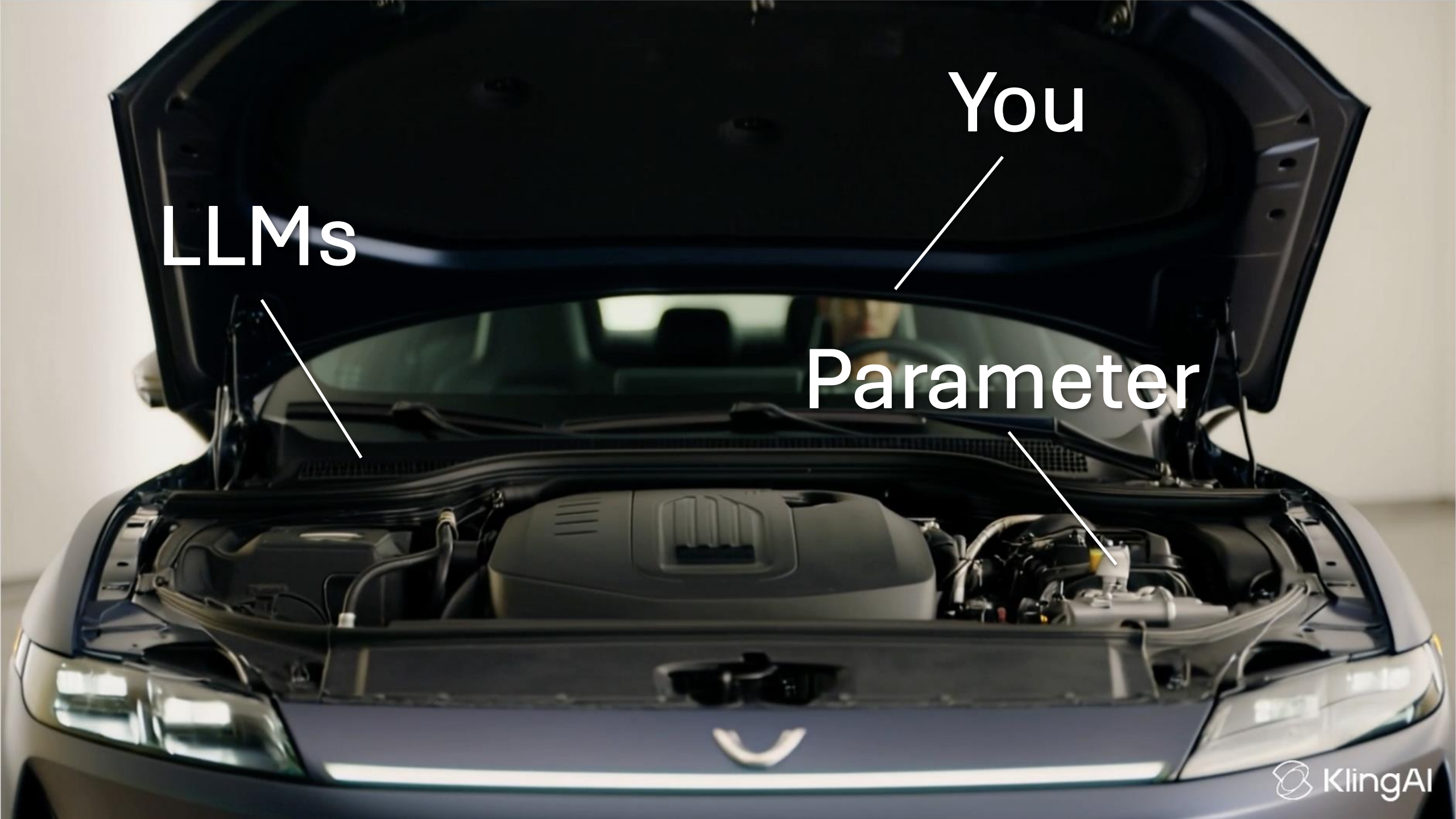
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KlingAI


?
KlingAI


Washer fluid reservoir
KlingAI


You
LLMs
Parameter
KlingAI

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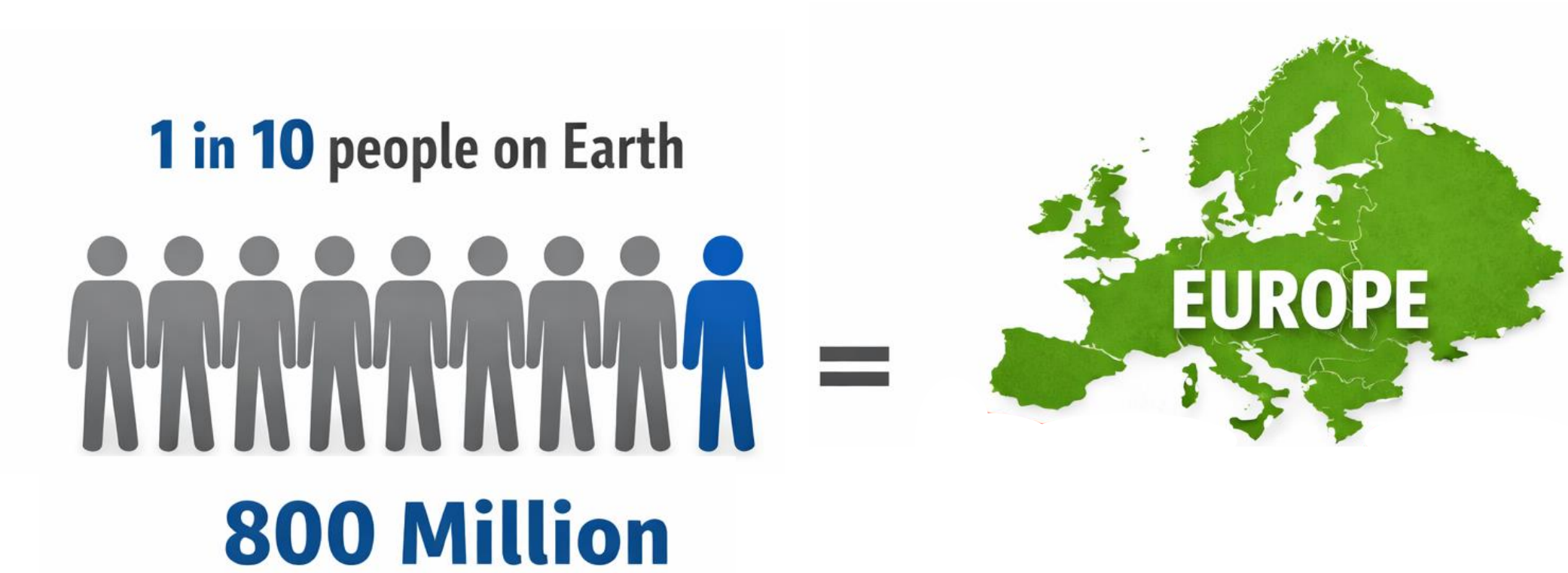
How many weekly active users does ChatGPT have?

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How many weekly active users does ChatGPT have?

2024



AI-generated image

ChatGPT Statistics Q3 2026: 1.2 Billion Weekly Users

The current global population is approximately 8.3 billion people as of October 2026
https://www.worldometers.info/world-population/

https://technologychecker.io/blog/chatgpt-statistics

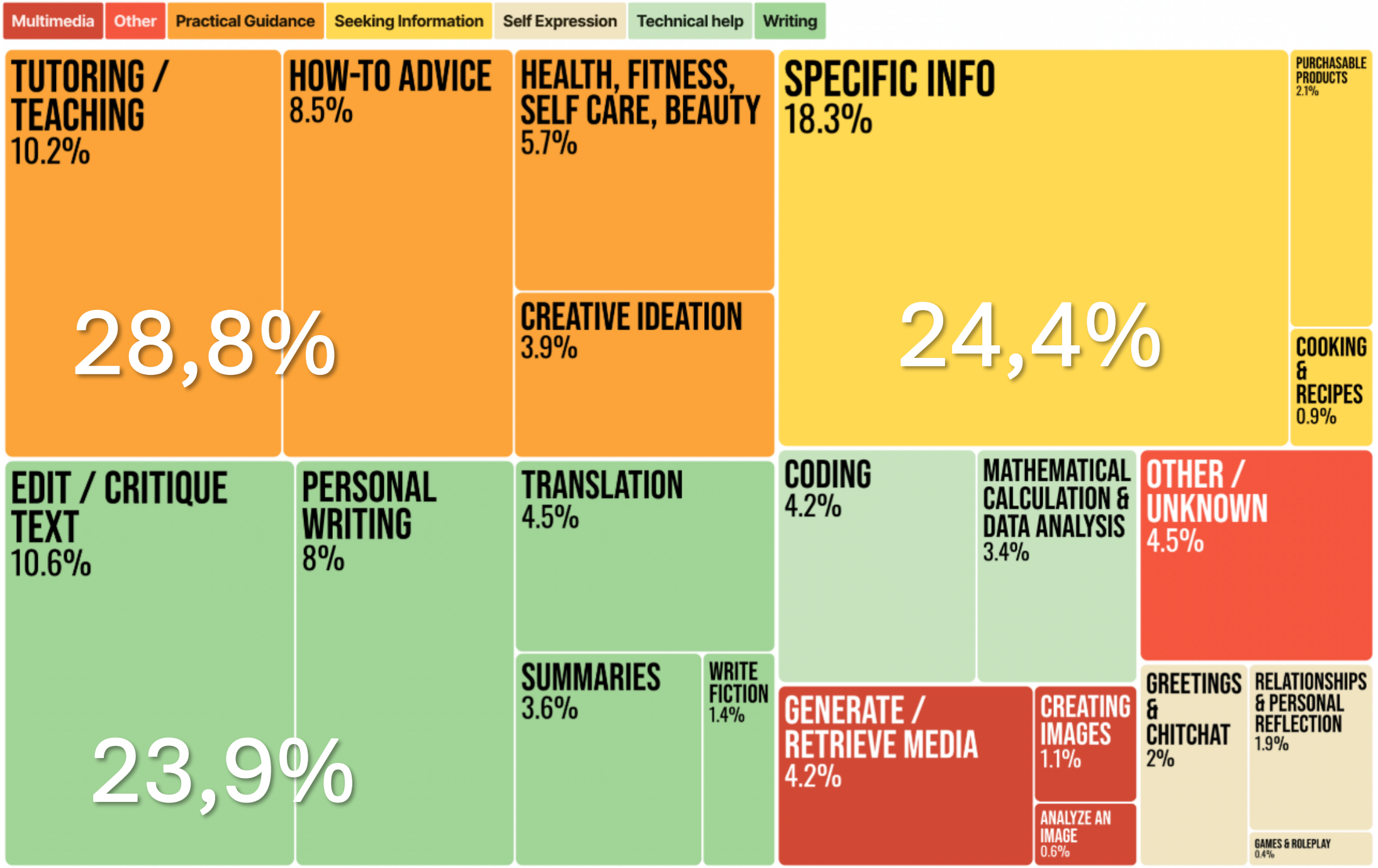
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What do you use ChatGPT for?

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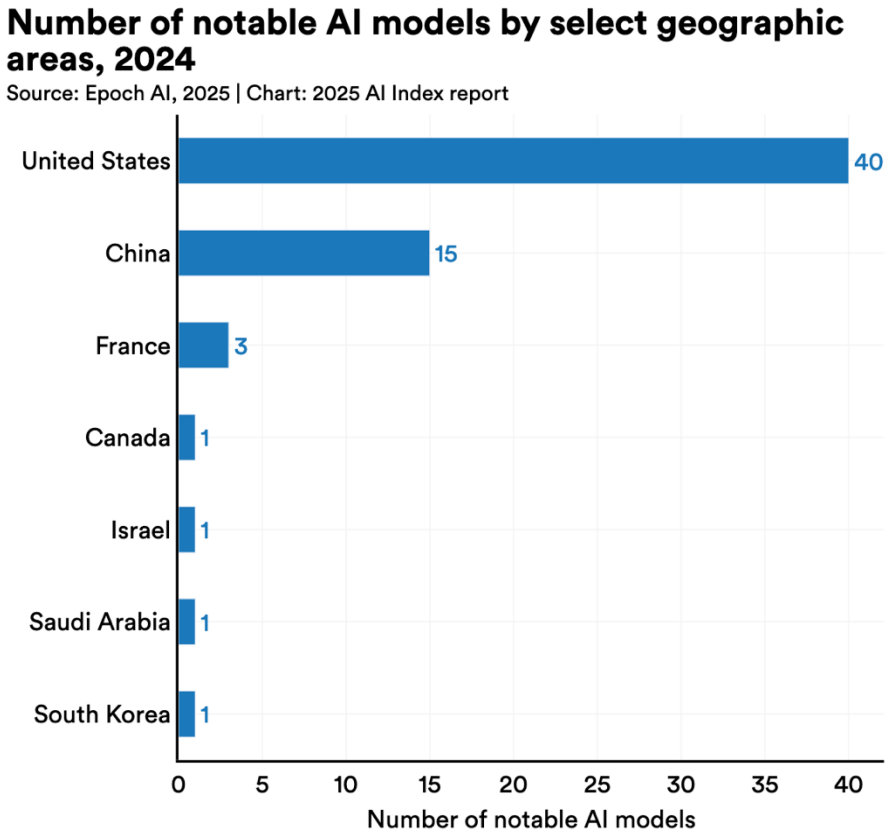
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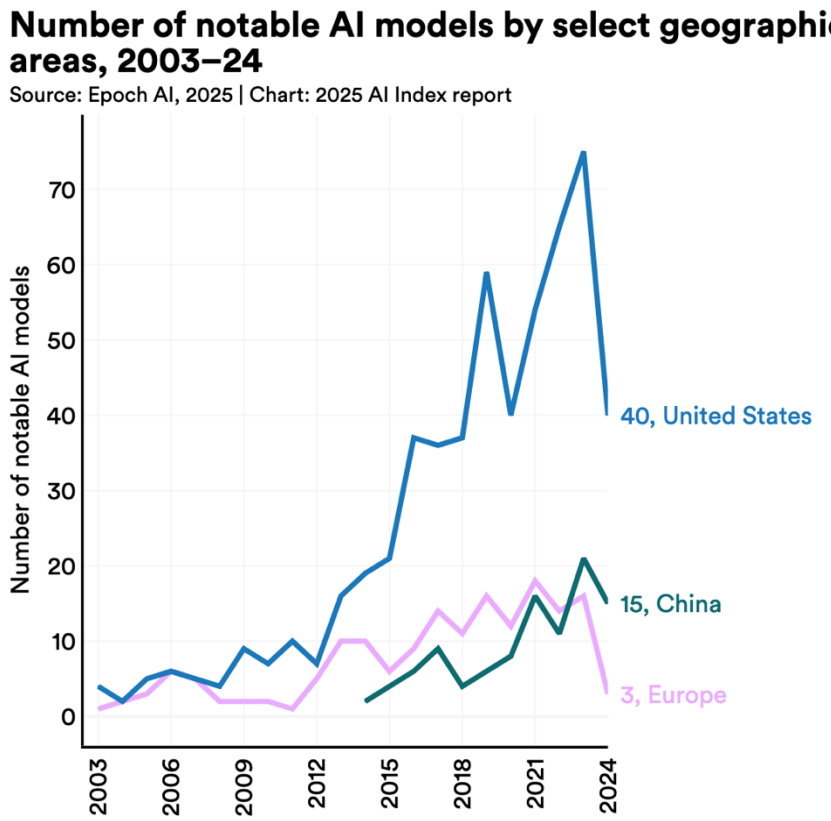


David McCandless **Information is Beautiful**

source: OpenAI

The Geographical Disparity of AI Innovation





The Geographical Disparity of AI Innovation

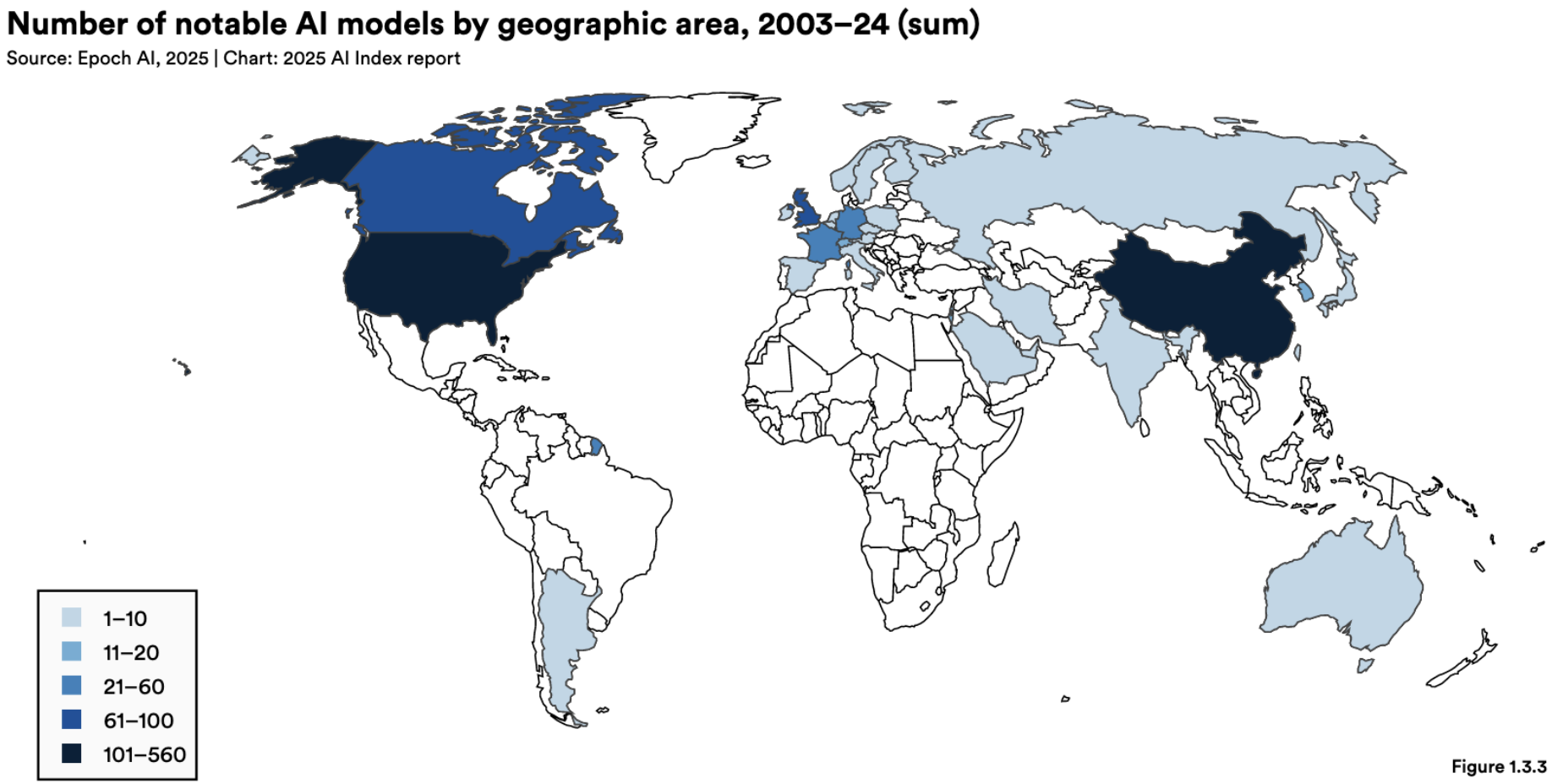
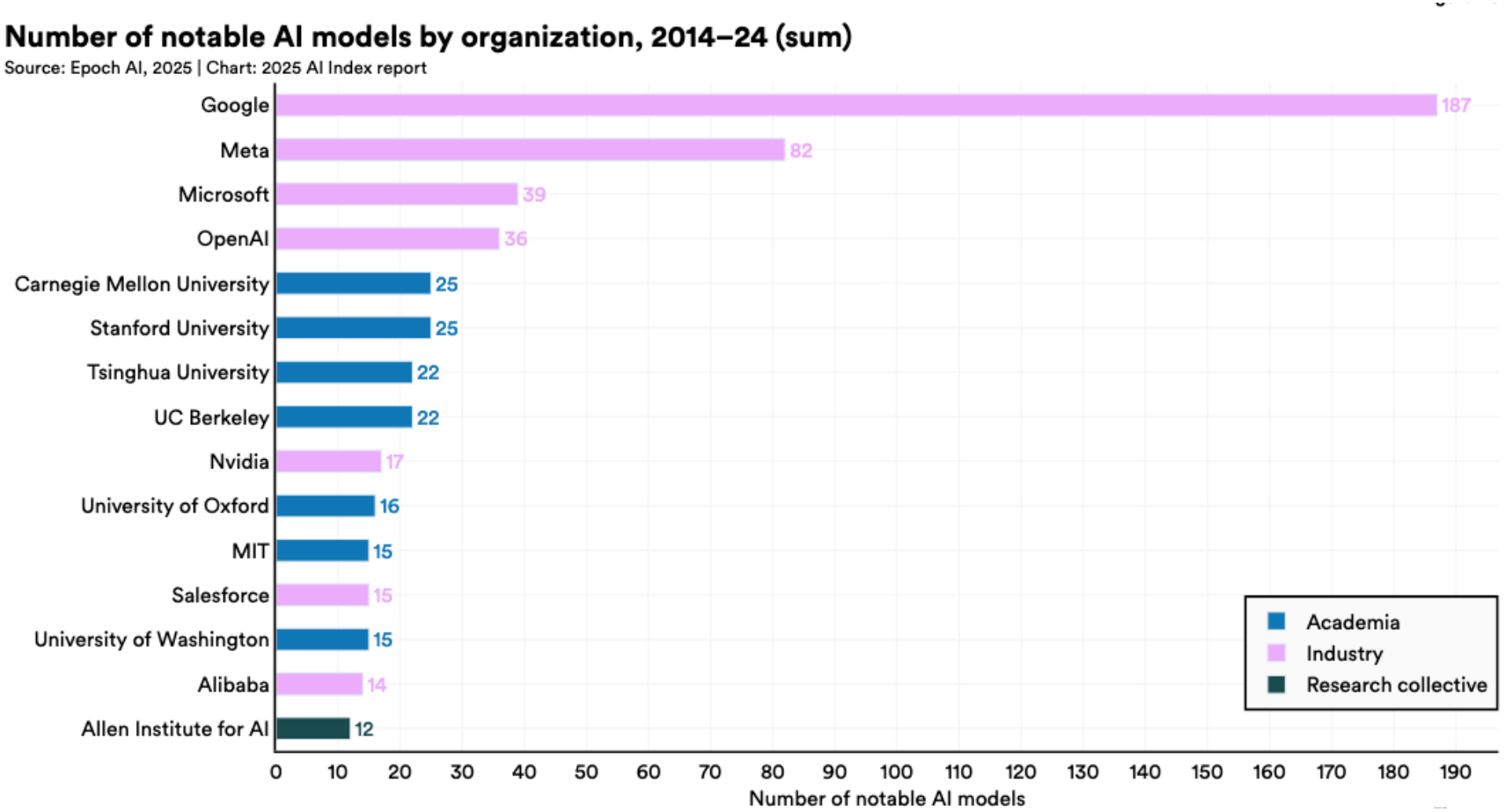


Figure 1.3.3

The Geographical Disparity of AI Innovation



The cost of training

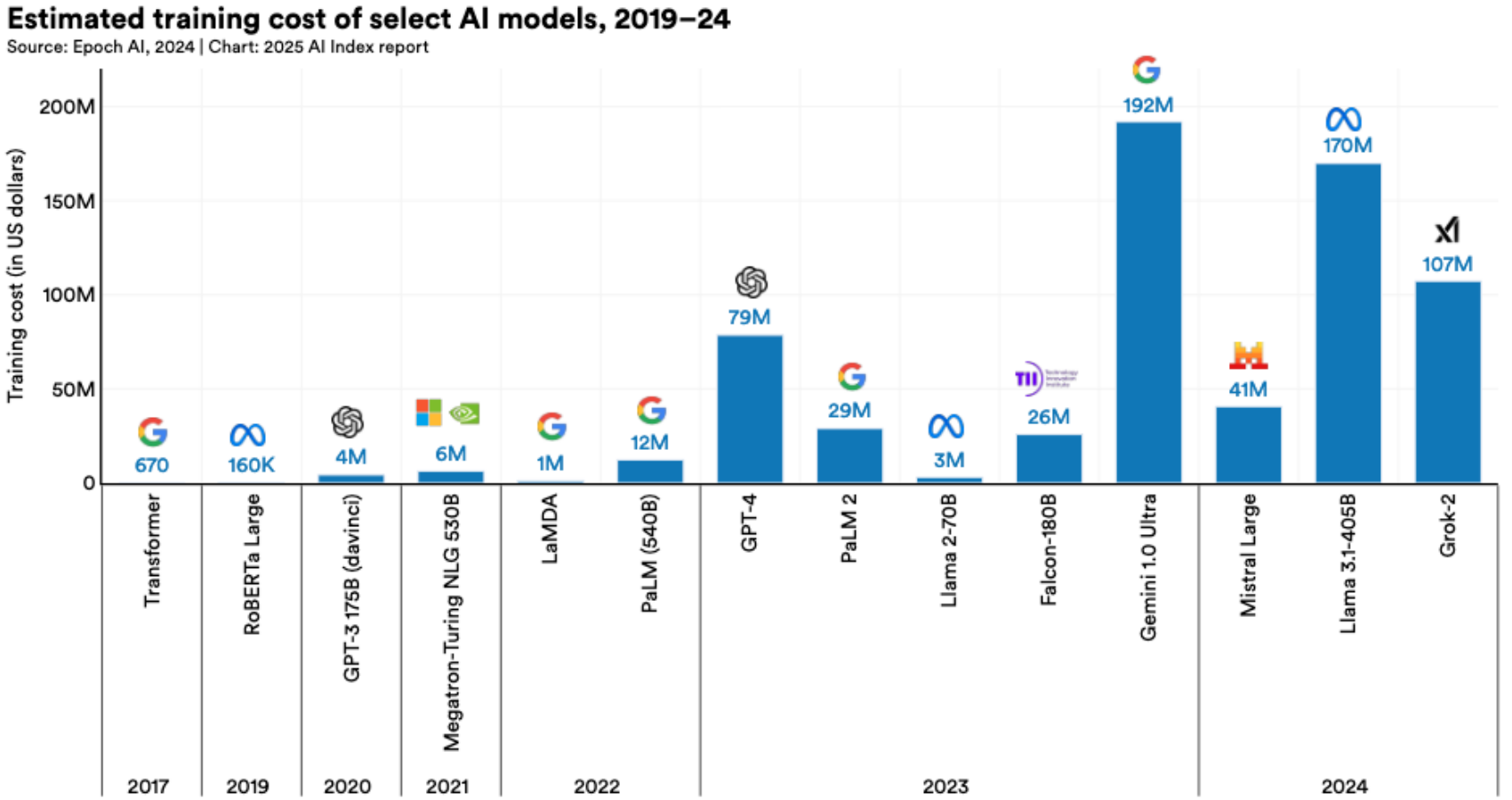


Figure 1.3.24

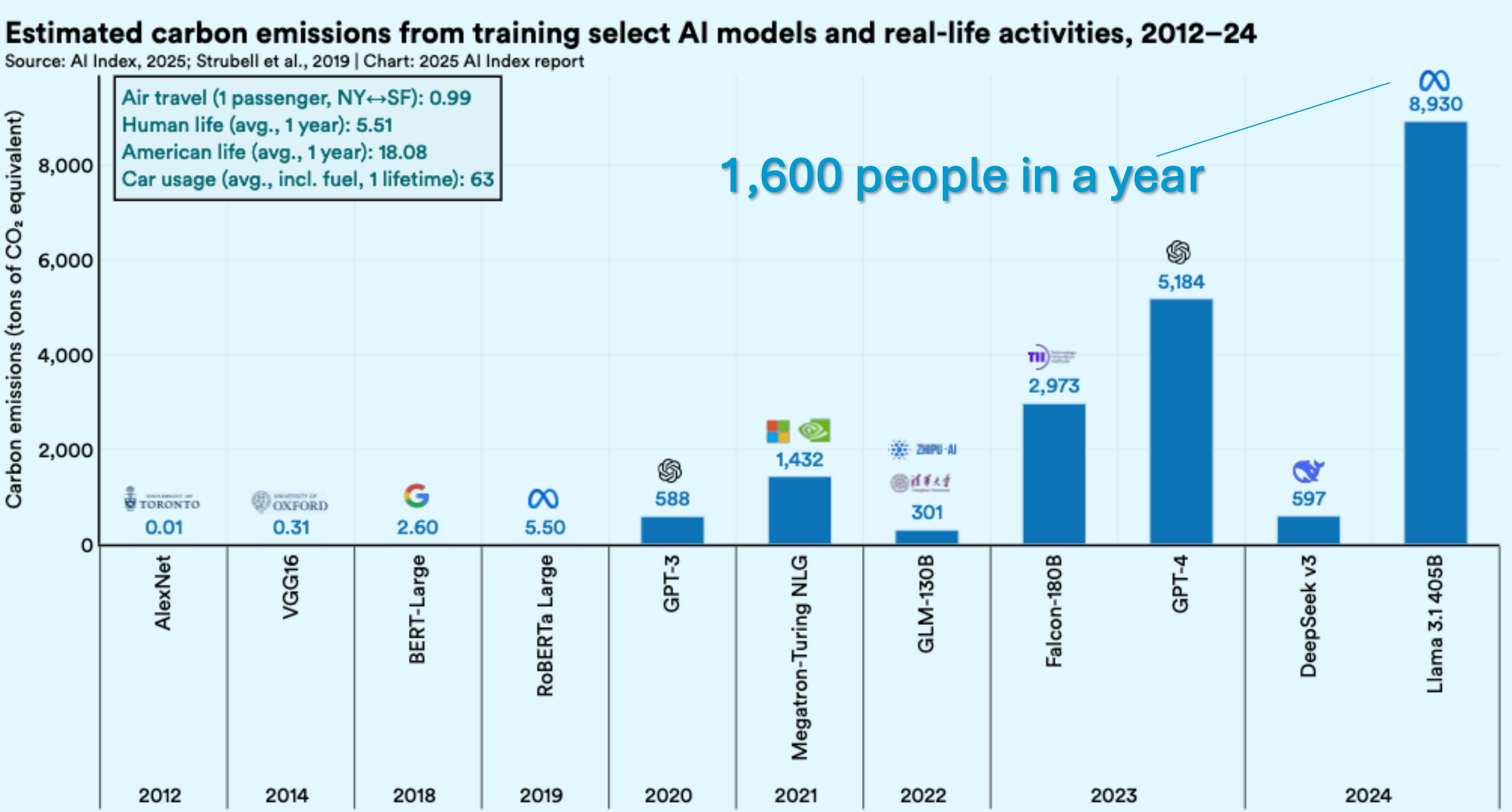

Estimated carbon emissions from training select AI models and real-life activities, 2012–24
Source: AI Index, 2025; Strubell et al., 2019 | Chart: 2025 AI Index report
Air travel (1 passenger, NY↔SF): 0.99
Human life (avg., 1 year): 5.51
American life (avg., 1 year): 18.08
Car usage (avg., incl. fuel, 1 lifetime): 63
1,600 people in a year
Carbon emissions (tons of CO₂ equivalent)
0
2,000
4,000
6,000
8,000
0.01
0.31
2.60
5.50
588
1,432
301
2,973
5,184
597
8,930
AlexNet
VGG16
BERT-Large
RoBERTa Large
GPT-3
Megatron-Turing NLG
GLM-130B
Falcon-180B
GPT-4
DeepSeek v3
Llama 3.1 405B
2012
2014
2018
2019
2020
2021
2022
2023
2024

Figure 1.4.7

All of this has been possible thanks to…

Natural Language Processing



Human language, the most **natural** to us.



Algorithms! But also Maths, coding, engineering, hardware, investments…

First, let’s try to define language

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Which language do you feel most comfortable in speaking?

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Language

A structured system of communication that consists of grammar and vocabulary to convey meaning in spoken or signed forms

(Wikipedia)

Language

A structured system of communication → What it does

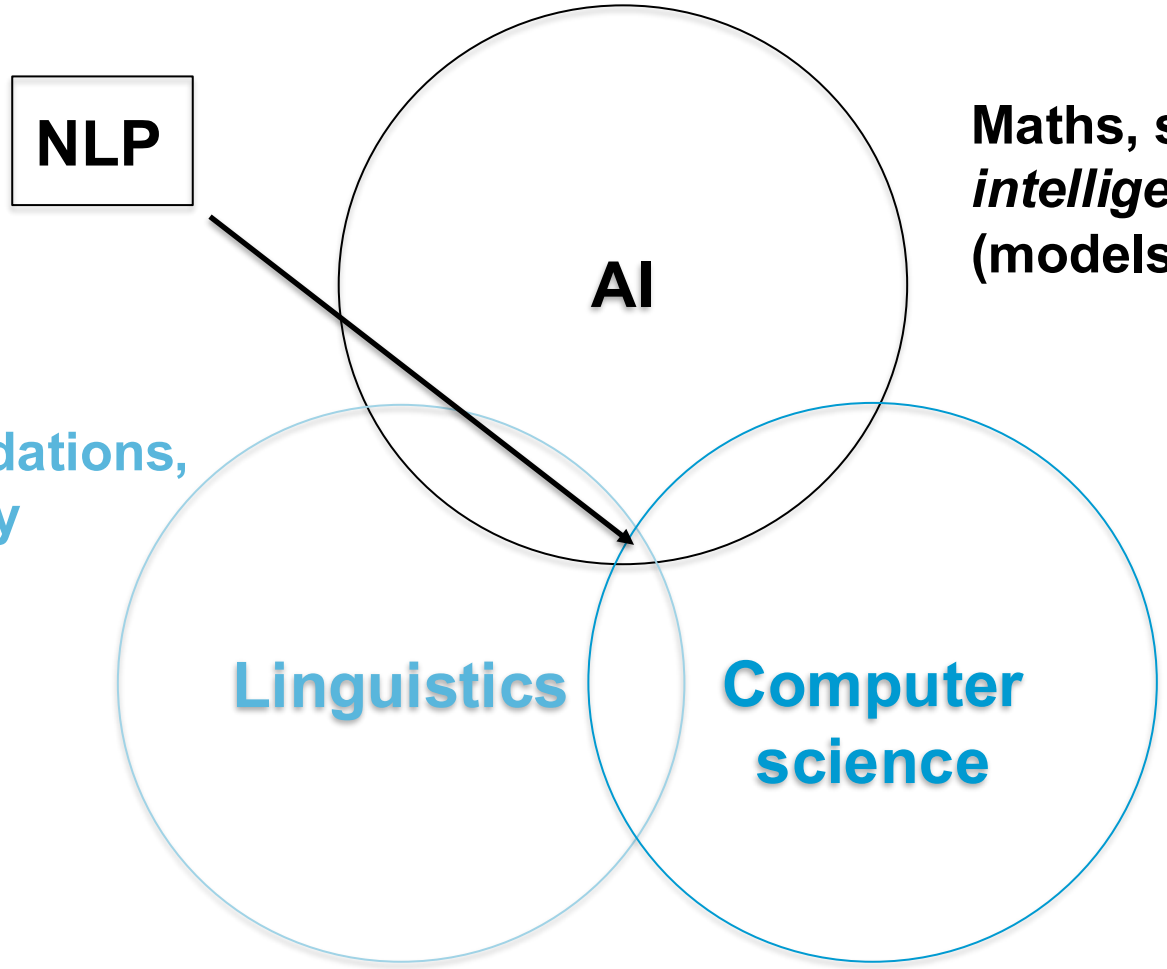
that consists of grammar and vocabulary → Entities

grammar → Rules

to convey meaning in spoken or signed forms → How

(Wikipedia)

Natural Language Processing

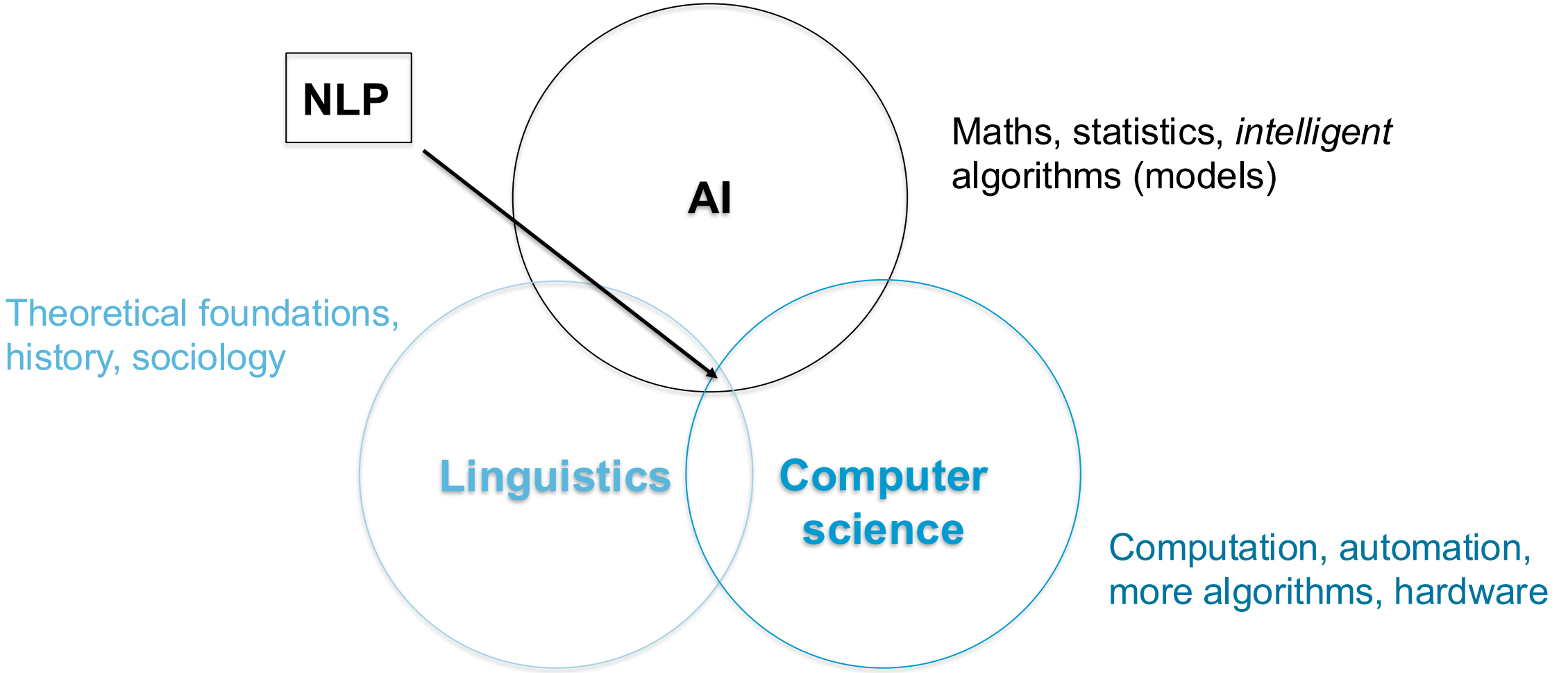


Maths, statistics, *intelligent* algorithms (models)

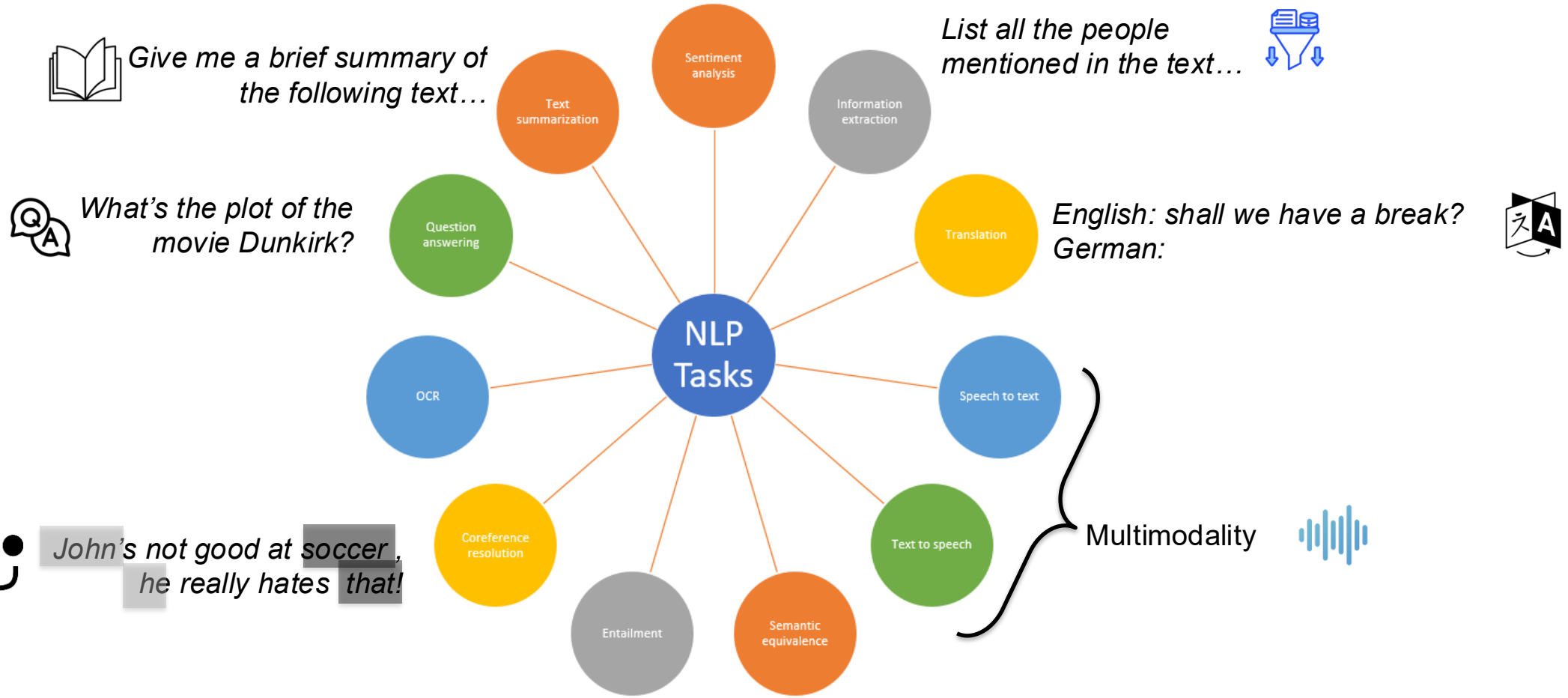
Theoretical foundations, history, sociology

Computation, automation, more algorithms, hardware

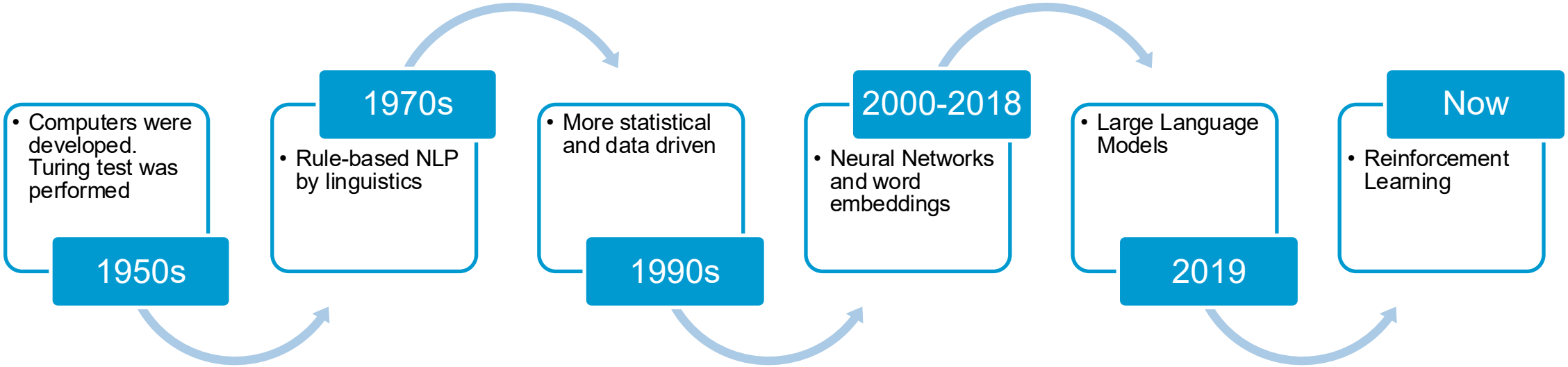
Natural Language Processing



What could we want NLP to do?



History of NLP





Credit: Accelerate Cambridge

What is a large language model?

A language model is a high-parameter model that is trained on massive amounts of text with the end goal of some kind of text generation.

- Not all language models have the same architecture
- Not all language models are built for next-token prediction
  - GPT is a next token predictor
  - BERT is a masked-language model
  - Not all language models are chatbots

Introduction to LLMs

Some definitions …

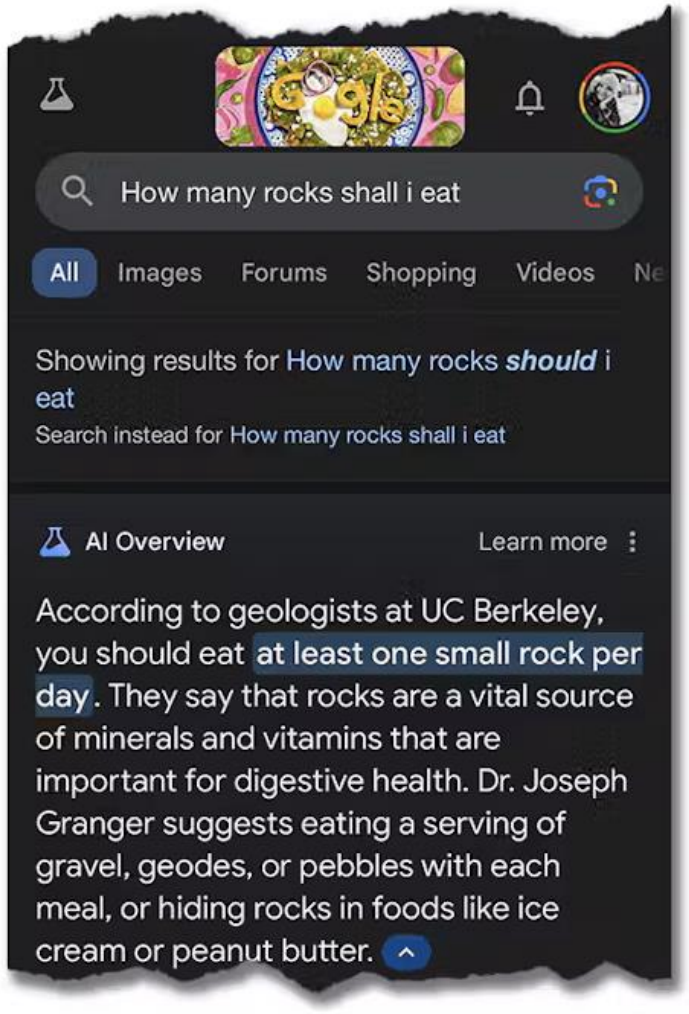
Model The thing you train and use to generate text.
Architecture The high-level structure of the model.
Parameters (or weights) The internal knobs and dials that get altered during the course of training.
Context The input to the LLM.
Embedding A high-dimensional representation of text.
Token The numerical representation of a single piece of text.
Pretraining The first stage of training the LLM, and usually the most expensive.
Finetuning Training the LLM to do a particular task.
Inference The act of calling the model for text generation.
Quantization Reducing the size of a model.
Source The availability of the model (can be open or closed).
Modality Defined by the type of data the model is trained on (e.g. text or images).
API (Application Programming Interface) The means by which you can interact with a model.

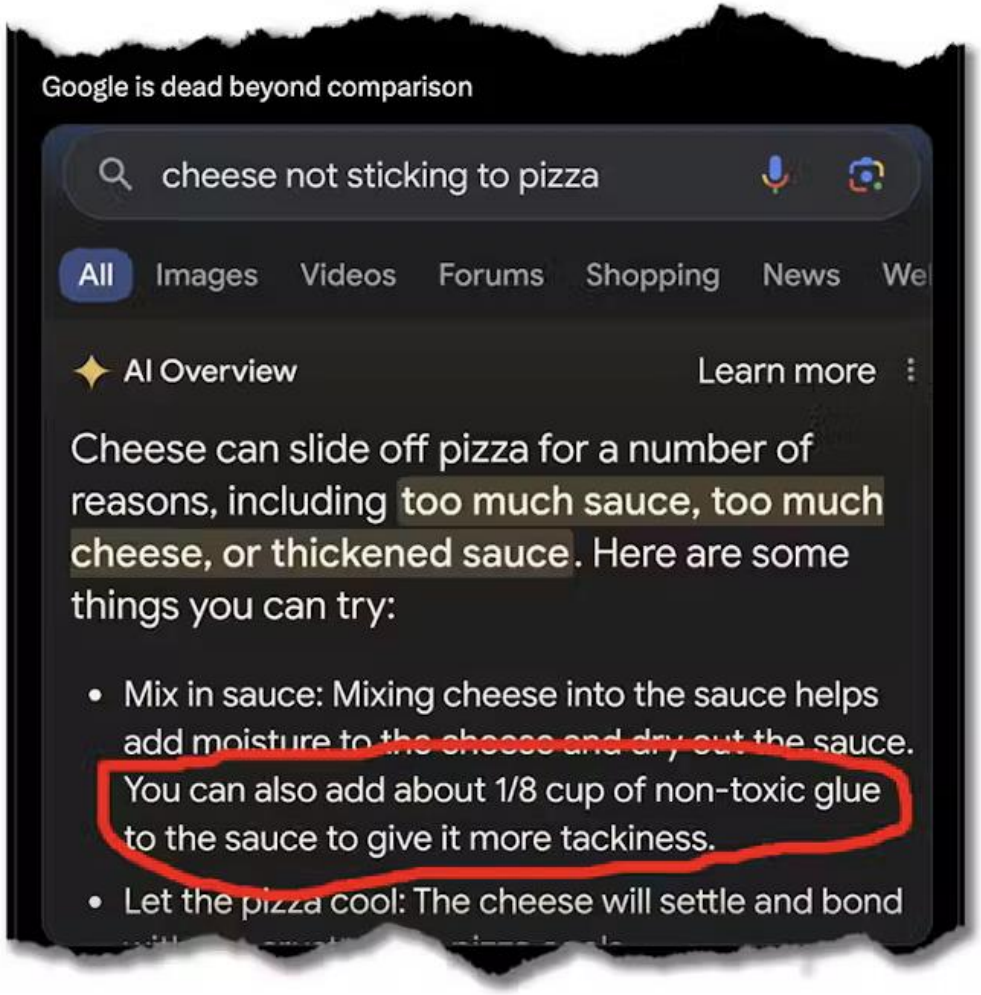
Why social scientists care

- Politics happens in text: speeches, manifestos, press releases, social media posts
- There is far more of it than anyone can read
- NLP lets us measure things like tone, topics, populism or polarization at scale

- **Example:** Do populists speak more simply than other politicians? One study measures this in 30 years of Bundestag speeches (session 4).

But: measurements can be wrong. Part of this course is learning when to trust them.





https://www.unsw.edu.au/newsroom/news/2024/05/eat-a-rock-a-day-put-glue-on-your-pizza-how-googles-ai-is-losing-touch-with-reality

What we have covered so far

What do you know about LLMs…?

What can you use them for?
What are potential problems?
What do you use them for?

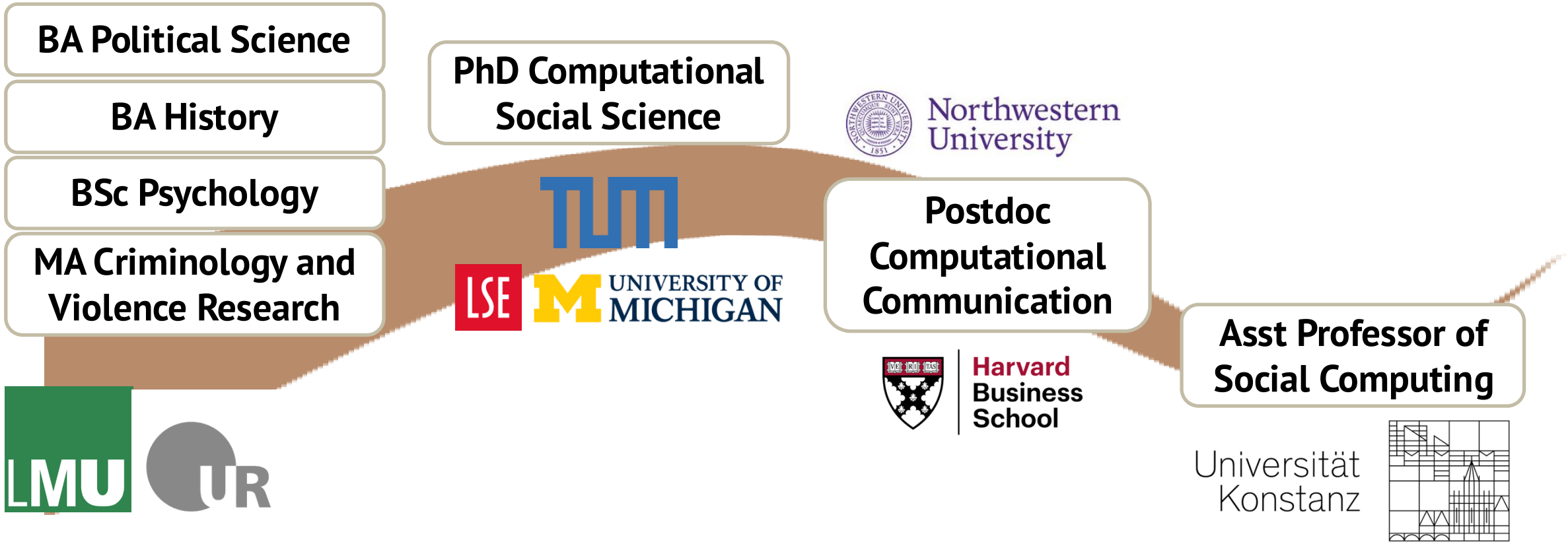
The rest of the course:

How do they work?

Questions?

Who am I?

How I came to Konstanz



Research Focus

DETECTING TRAUMA IN GENOCIDE WITNESS TESTIMONIES

HATE SPEECH AND RADICALIZATION

PERCEPTIONS OF HATE AND INCITEMENT TO VIOLENCE



Your turn!

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Which program are you in?

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Have you ever written code before?

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What’s your favorite spot in Konstanz?

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This course

What you will learn

- How NLP methods and LLMs work, from counting words to transformers
- How to apply basic text analysis to political and social science data, in Python
- How to critically evaluate research that uses these methods

No prior programming experience is required.

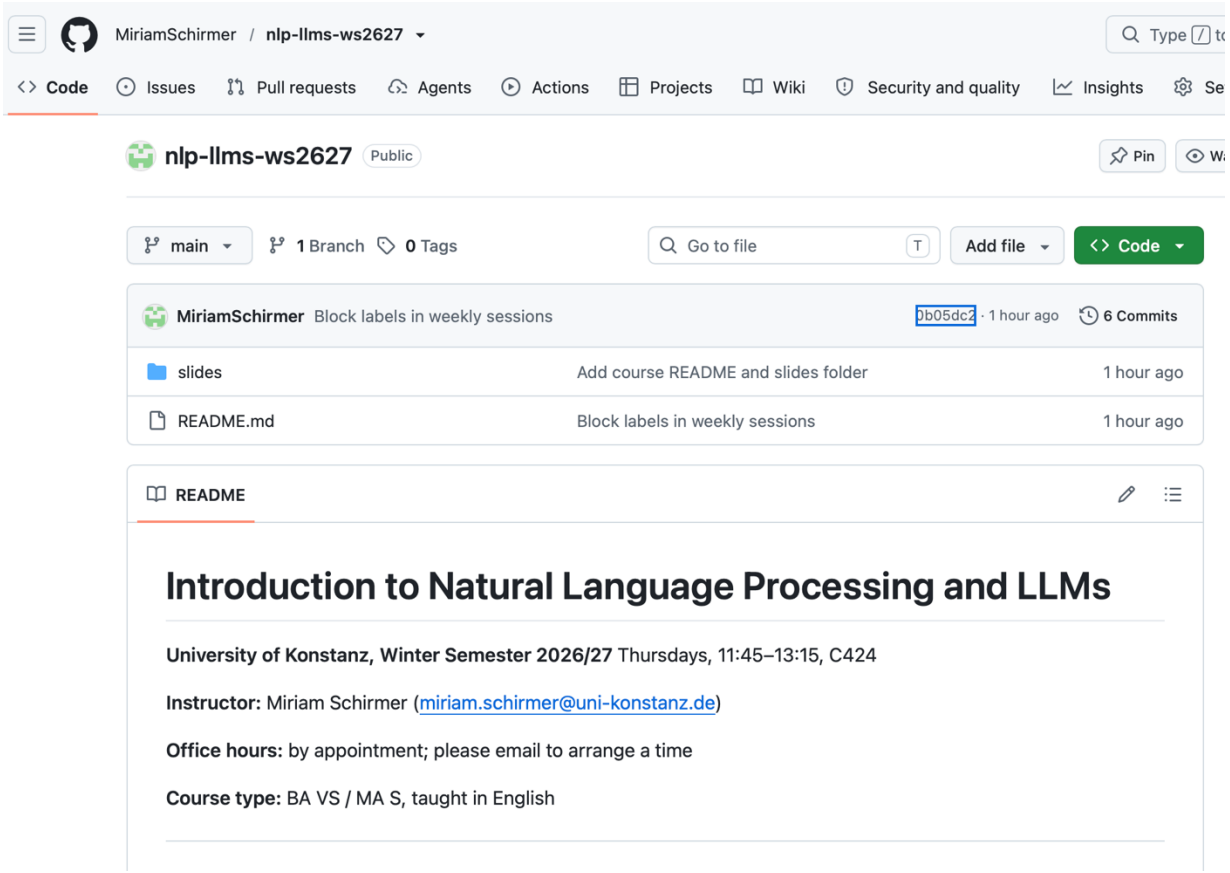
Course Structure

- **Block 1:** Text as Data
- **Block 2:** Learning from Text
- **Block 3:** Embeddings and Transformers
- **Block 4:** LLMs as Research Tools

Each block: theory, then practical, then paper session

- Theory: concepts and readings. Practical: Python notebooks in Google Colab. Papers: you present applied research.

Schedule & Materials



Tools

- **ILIAS:** announcements – please register!
- **Google Colab:** Python in your browser, nothing to install.
- **Slido:** short questions during class, anonymous, no account needed
- **Course repository on GitHub:** notebooks and materials (link follows)

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What would you most like to be able to do after this course?

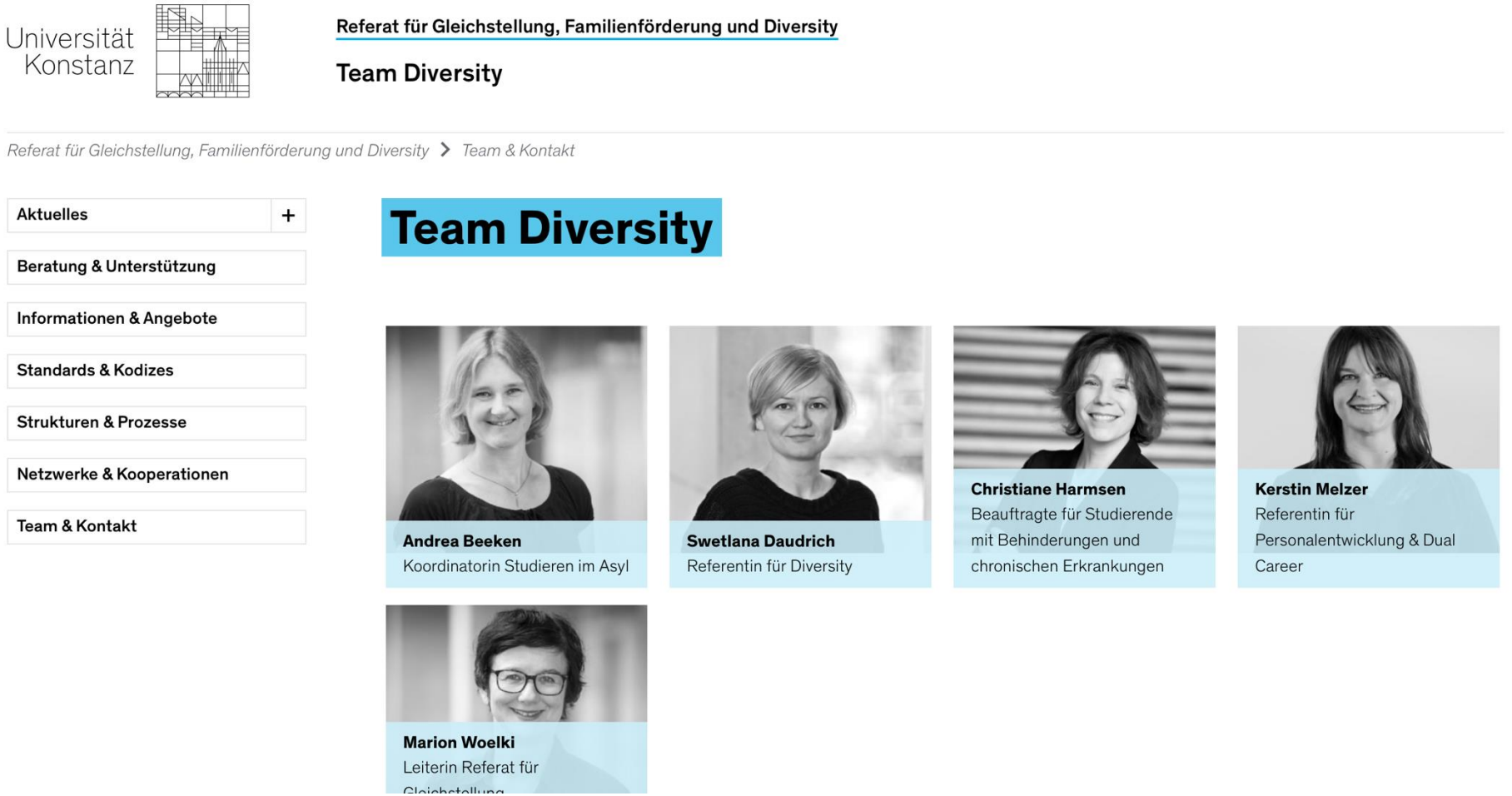
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Before next week (29 October)

- **Register in ZEuS and ILIAS**
- **Read:** Jurafsky & Martin, ch. 2 "Words and Tokens" and Benoit (2020) "Text as Data: An Overview" – pick one thing that you liked / didn't know before
- **Sign up** for a paper presentation

- All readings are linked in the syllabus.



Offers the Diversity Office: https://www.uni-konstanz.de/gleichstellungsreferat/gleichstellung-in-wissenschaft-und-studium-2-1-1-1/

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Questions?
Thank you!

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Credits: Parts of this slide deck have been adapted from slides created by Jurafsky & Martin, Debora Nozza, and Julia Mendelsohn.